UniSphere Rquirements

UniSphere Requirements:

1. Functional Requirements

A. Authentication & Security

ID	Requirement
FR-A1	Users shall log in using email/password credentials.
FR-A2	The app shall support biometric authentication (fingerprint/face) to unlock the app after the first login.
FR-A3	Session tokens or credentials shall be stored securely using local secure storage mechanisms.
FR-A4	The system shall provide logout functionality and proper session termination.

B. Networking & Offline

ID	Requirement
FR-B1	The app shall fetch data from a REST API (e.g., JSONPlaceholder or custom endpoint).
FR-B2	The system shall support **Online Mode** for fetching and refreshing live data.
FR-B3	The system shall support **Offline Mode**, allowing users to browse cached content without connectivity.
FR-B4	The UI shall display clear states: **Loading**, **Error**, and **Offline Banner**.

C. Data & Local Persistence & Storage

ID	Requirement
FR-C1	Data types managed shall include **Announcements**, **Events**, and **Timetable**.
FR-C2	User settings (theme, language, notification flags) shall be managed via `SharedPreferences` and `FlutterSecureStorage`.

FR-C3	Structured cached content for offline use shall be persisted using a local database.
FR-C4	The app shall provide File I/O capability to export user schedules as JSON files.

D. Navigation & User Interface

ID	Requirement
FR-D1	The app shall provide **5 main screens**: Login, Home (Dashboard), Announcements, Events, and Settings.
FR-D2	Navigation shall be implemented using **named routes** or `onGenerateRoute` route settings.

E. Device Features & Permissions

ID	Requirement
FR-E1	The app shall request **runtime permissions** for device features and handle denials gracefully.
FR-E2	**Camera/Gallery**: Users shall attach photos to event notes.
FR-E3	**Location**: The app shall display the user's position relative to campus Points of Interest (POIs).
FR-E4	**Sensors**: The app shall utilize the accelerometer for simple interactions (e.g., shake-to-refresh).
FR-E5	**Bluetooth/NFC**: A conceptual demo for hardware-based campus interactions shall be included.

F. Notifications & Background Execution

ID	Requirement
FR-F1	The app shall support **local notifications** to schedule reminders (e.g., 10 minutes before a class).
FR-F2	The system shall implement **background tasks** for periodic data refreshes (e.g., announcements).
FR-F3	At least **one scheduled local notification** and **one tap action** (e.g., deep link to event detail) shall be implemented.

2. Non-Functional Requirements

A. Architecture & Code Quality

ID	Requirement
NFR-A1	The app shall be built using **Flutter stable** with **Dart null safety**.
NFR-A2	Architecture shall follow **Clean Architecture-lite** (Data/Domain/Presentation) or **MVVM with Repositories**.
NFR-A3	State management shall use **BLoC**, **Riverpod**, or **ValueNotifier**.
NFR-A4	Code shall demonstrate meaningful naming, separation of concerns, and robust error handling.
NFR-A5	The code shall adhere to SOLID principles.

B. Security

ID	Requirement
NFR-B1	All sensitive data shall use **secure storage** and follow OWASP mobile security considerations.
NFR-B2	Biometric authentication shall integrate with platform-specific secure APIs (e.g., `local_auth`).

C. Performance & Reliability

ID	Requirement
NFR-C1	The app shall work **reliably online and offline** (offline-first reliability).
NFR-C2	Frame rendering and list scrolling shall be optimized to maintain smooth UI performance (target 60 FPS).
NFR-C3	Image caching and widget rebuild optimization shall be applied where applicable.
NFR-C4	The system shall be optimized to minimize unnecessary widget rebuilds, improve list rendering efficiency, and apply image caching, as validated through performance profiling tools.

D. Usability & Accessibility

ID	Requirement
NFR-D1	The UI shall be accessible, featuring readable typography, high-contrast modes, and screen-reader compatibility.
NFR-D2	Navigation shall be intuitive with consistent visual hierarchy and feedback.